Future Control Moves Memory: A Long-Horizon Moved-Bottleneck Diagnostic for Synthetic Agents

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Abstract

This paper asks a narrow question about finite neural agents: when one early state variable becomes future-critical for a delayed decision, does the agent's memory geometry become specifically sensitive to that variable, and does that sensitivity move when the critical variable moves?

We introduce the **long-horizon moved-bottleneck diagnostic**. Four early clue slots are matched for salience and frequency. One slot is selected as the future-critical bottleneck, moved across registered positions, and queried only after a long delay. The primary metric is not just final accuracy; it is the relative displacement of the final hidden state under single-slot bit flips. The registered gate requires the final-state sensitivity peak to follow the moved critical slot while remaining absent in a visible-control condition.

The result is positive across a ladder of increasingly agent-like synthetic regimes. A transformer on Modal `L4` solves the base delayed-memory task, survives longer horizons through sequence length 384, and preserves the moved bottleneck through closed-loop external-state handoff, repair after tool error, parsed structured calls, multi-field tool schemas, stochastic API failures, an 8-slot larger argument namespace, an alias-rich argument surface, parser-facing text arguments, fixed-length generated JSON strings, and autoregressively decoded JSON-like action strings. Across the confirmed Modal reports, the relevant bottleneck groups reach 1.000 closed-loop accuracy and positive moved-slot specificity, while visible controls keep specificity near zero.

The first prompt-level transfer produced a useful controlled strong negative at one hidden site: `Qwen/Qwen2.5-0.5B-Instruct` emits valid parser-scored JSON actions and passes the behavioral gates, but final-prompt hidden specificity is not confirmed. A follow-up localization sweep resolves that ambiguity. Across `Qwen/Qwen2.5-0.5B-Instruct`, `Qwen/Qwen2.5-1.5B-Instruct`, and `HuggingFaceTB/SmolLM2-1.7B-Instruct`, all behavior controls pass and 17 registered hidden sites pass. The strongest signal appears at final generated JSON tokens, with additional late/final prompt-token sites passing for Qwen 1.5B.

The diagnostic now also has a black-box API surface. A dependency-free evaluator emits JSONL rows and scored summaries for fixture, Gemini, Anthropic, OpenAI, and OpenAI-compatible providers. `gemini-3.1-flash-lite` passes the registered prompt-family behavior suite and a small external-stress suite covering wider slot sets, longer filler gaps, and API-dispatch prompt framings. This is behavioral evidence only, not hidden-state evidence.

The allowed claim is deliberately modest: in this synthetic neural-agent setting, **future control relevance can move finite memory-state and tool-commitment sensitivity**. This is not a production-agent reliability or consciousness claim. Its value is as a cheap, reproducible diagnostic for whether an agent's internal state, generated action trajectory, or black-box tool-call behavior tracks the variables that will later control action.

1. Motivation

Many long-horizon agent evaluations ask whether an agent eventually gives the right answer. That is necessary but weak. A system can succeed by a shortcut, by storing everything uniformly, by using local query cues, or by relying on a privileged final token. The moved-bottleneck diagnostic tests a sharper property: whether the agent's internal memory metric follows the variable that will matter later.

This matters for three fields.

1. **Interpretability:** it gives a controlled way to ask where future-relevant information lives in a model's hidden state.
2. **Agent evaluation:** it tests delayed commitment and repair, not just answer selection at the final step.

3. **Safety and monitoring:** it suggests a cheap diagnostic for variables that silently become control bottlenecks in long-horizon systems.

The experiment is intentionally synthetic. That is a feature, not a defect, for this stage: the task isolates memory transport, critical-variable movement, and tool-interface hardening before introducing the confounds of natural language and real APIs.

2. Diagnostic Design

Each episode contains matched early clues:

```
slot_0: bit b0
slot_1: bit b1
slot_2: bit b2
slot_3: bit b3
...
delayed query
```

In the `bottleneck` condition, the correct final action is the bit at one registered critical slot. The critical slot is moved across runs. In the `visible_control` condition, the final answer is visible at the query token and the early clues are matched distractors.

For each trained cell, the metric intervention flips one early clue bit at a time and measures the final hidden-state displacement. Slot densities are z-scored within model. The primary specificity statistic is:

```
z_density(critical_slot) - mean z_density(noncritical_slots)
```

The core gates are:

- behavior: bottleneck accuracy at least 0.90;
- metric transport: bottleneck memory-specificity CI lower bound above zero;
- rank: the critical slot ranks above chance among registered slots;
- visible-control null: visible-control specificity remains below 0.50.

The tool-interface regimes add gates for committed slot/value accuracy, schema validity, parsed action validity, repair behavior, stochastic failure handling, and no-op behavior after successful first calls.

3. Evidence Ladder

The experiment was advanced through a sequence of regimes. Each regime keeps the moved critical slot and visible-control null, then changes one surface of the agent's interaction with future-relevant information.

Regime	What changed	Cells	Key signal	Report key
Base moved bottleneck	Delayed query over four matched clue slots	64	Bottleneck final accuracy 1.000; memory specificity +2.309; visible null	`base`
Horizon stress	Delay lengths 128, 256, 384	96	All lengths preserve final accuracy 1.000 and moved-slot specificity	`horizon`
Closed-loop tool handoff	Model commits slot/ value to recover external state	32	Tool bottleneck final accuracy 1.000; slot/ value commitment 1.000	`commit`

Repair bottleneck	First tool attempt receives an error; model must repair	48	Repair condition final accuracy 1.000; repair slot/value 1.000	`repair`
Structured tool call	One parsed JSON-like action token replaces direct heads	48	Direct and repair structured calls pass schema and parsed-slot gates	`structured`
Multifield schema	Separate opcode, slot, and value fields	48	Multifield direct/repair groups pass field, schema, and parsed-value gates	`multifield`
Stochastic tool failure	First-call success/ failure sampled per episode	32	Failed repair and success no-op both 1.000; failure rate 0.506	`stochastic`
8-slot stochastic	Larger argument namespace and longer sequence	64	8-slot stochastic gates pass; memory specificity +3.023	`8slot`
Alias argument surface	Three equivalent aliases per canonical slot	32	Alias parsed slot/value, failed repair, and success no-op all 1.000	`alias`
Text argument surface	Parser-facing phrases such as `second clue` replace alias IDs	32	Text parsed slot/value, failed repair, and success no-op all 1.000	`text`
Generated JSON surface	Model emits a fixed-length JSON-like token sequence	32	Generated sequence/ schema, parsed slot/ value, and repair gates all 1.000	`generated_json`
Autoregressive JSON surface	JSON action is decoded token-by-token from commit state	32	Greedy decoded sequence/schema, parsed slot/value, and repair gates all 1.000	`autoregressive_json`
Prompt JSON transfer	Pretrained open model emits parser-scored JSON text	64	Controls and behavior pass; hidden specificity CI crosses zero	`prompt_json_transfer`
Prompt JSON hidden localization	Multi-model, multi-layer, multi-token hidden-site grid	192 behavior cells + 576 hidden rows	Controls and behavior pass for three models; 17 hidden sites pass	`prompt_json_hidden_localization`
Prompt JSON fixed-action localization	Teacher-forced constant assistant actions under slot flips	192 behavior cells + 960 hidden rows	Controls and behavior pass; 24 hidden sites pass after generated action tokens are held fixed	`prompt_json_fixed_action_localization`
Prompt JSON causal patch	Patch base hidden state into critical-slot-flipped prompt before JSON value token	1,536 causal-patch rows	Value-prefix patch sites pass in all three model families; prompt-final sites do not pass	`prompt_json_causal_patch`
Prompt-family causal patch	Repeat causal patch over standard, compact, and audit-checklist prompt framings	4,608 causal-patch rows	All 9 family/model pairs are causally ready and patch-pass	`prompt_json_prompt_family_causal_patch`

API black-box prompt-family	One-command provider evaluator over parser-scored JSON actions	96 Gemini rows	Gemini 3.1 Flash-Lite passes all registered prompt-family behavior cells	`api_blackbox_gemini`
API black-box external stress	Wider slots, longer gaps, retrieval and dispatch prompt framings	80 Gemini rows	All 20 stress/family behavior cells pass	`api_blackbox_gemini`

All confirmed sweeps used Modal `L4`, not H100/H200. The timeout-based conservative spend guard for the listed confirmed reports is under USD 190 in aggregate; individual confirmed passes used guards of USD 1.08 to USD 17.26. Actual runtime was much lower than the timeout budget, with the latest autoregressive JSON pass averaging 15.08 seconds per remote cell. The prompt-level transfer used one `L4` container for all 64 logical cells, with a conservative timeout guard of USD 1.08 and a 212.0 second remote runtime. The prompt-level hidden-localization sweep used three parallel `L4` model jobs, 192 behavior cells, 576 hidden rows, and a conservative timeout guard of USD 6.47. The fixed-action localization counterfactual used the same three parallel `L4` model jobs, 192 behavior cells, 960 hidden rows, and the same conservative timeout guard of USD 6.47. The causal-patch pass used three parallel `L4` model jobs, 1,536 patch rows, and the same conservative timeout guard of USD 6.47. The prompt-family causal-patch robustness pass used three parallel `L4` model jobs, 4,608 patch rows, and the same conservative timeout guard of USD 6.47. The black-box API runs used provider calls rather than GPUs: the Gemini prompt-family run used 144 API requests for 96 scored rows, and the external-stress run used 120 API requests for 80 scored rows.

Report keys map to committed summaries under `experiments/long_horizon_bottleneck/results/`: `base` = `modal_transformer_l4_8seed_2026_07_02.md`, `horizon` = `modal_transformer_l4_horizon_4seed_2026_07_02.md`, `commit` = `z_closed_loop_tool_commitment_l4_4seed_2026_07_02.md`, `repair` = `zz_tool_recovery_l4_4seed_2026_07_02.md`, `structured` = `zzz_structured_tool_call_l4_4seed_2026_07_03.md`, `multifield` = `zzzz_multifield_tool_schema_l4_4seed_2026_07_03.md`, `stochastic` = `zzzzz_stochastic_tool_failure_l4_4seed_2026_07_03.md`, `8slot` = `zzzzzz_stochastic_tool_failure_8slot_l4_4seed_2026_07_03.md`, and `alias` = `zzzzzzz_alias_argument_surface_l4_4seed_2026_07_03.md`, `text` = `zzzzzzzz_text_argument_surface_l4_4seed_2026_07_03.md`, `generated_json` = `zzzzzzzzz_generated_json_surface_l4_4seed_2026_07_03.md`, and `autoregressive_json` = `zzzzzzzzzz_autoregressive_json_surface_l4_4seed_2026_07_03.md`, and `prompt_json_transfer` = `zzzzzzzzzzz_prompt_json_transfer_l4_4seed_2026_07_03.md`, and `prompt_json_hidden_localization` = `zzzzzzzzzzzzz_prompt_json_hidden_localization_l4_4seed_2026_07_03.md`, and `prompt_json_fixed_action_localization` = `zzzzzzzzzzzzzz_prompt_json_fixed_action_localization_l4_4seed_2026_07_03.md`, and `prompt_json_causal_patch` = `zzzzzzzzzzzzzzz_prompt_json_causal_patch_l4_4seed_2026_07_03.md`, and `prompt_json_prompt_family_causal_patch` = `zzzzzzzzzzzzzzzz_prompt_json_prompt_family_causal_patch_l4_4seed_2026_07_03.md`, and `api_blackbox_gemini` = `zzzzzzzzzzzzzzzzz_api_blackbox_gemini_flash_lite_2026_07_06.md`.

4. Prompt-Level JSON Transfer and Hidden Localization

The autoregressive JSON pass completed the synthetic ladder: the trained model could decode parser-scored JSON-like action strings token by token under stochastic first-call failure and conditional repair. The next question was whether the same diagnostic transfers to a pretrained prompt-level model.

The prompt-level transfer used `Qwen/Qwen2.5-0.5B-Instruct` on one Modal `L4` container. The model saw natural-language memory records and had to emit one JSON object:

```
{"tool":"read_slot","slot":"second clue","value":0}
```

or:

```
{"tool":"noop"}
```

The confirmatory run used 64 logical cells: 4 conditions, 4 moved critical slots, and 4 seeds, with 8 episodes per cell. Format, visible-control, and short-horizon controls passed. The behavioral bottleneck also passed: closed-loop final accuracy was 0.977, first parsed slot/value were 0.984/0.977, failed-repair slot/value were 1.000/1.000, and success repair no-op was 1.000.

The full prompt-level gate nevertheless failed by the preregistered hidden metric. Critical-slot memory specificity was +0.695, but its 95% bootstrap interval was [-0.376, 2.080], so the lower bound was not above zero. The rank metric was above chance by mean rank percentile (0.594), but the primary specificity gate did not pass.

This is a controlled strong negative, not an interface failure. The model can solve the parser-scored prompt-level moved-bottleneck behavior, including stochastic repair, but this run does not show reliable hidden-state transport of the future-critical slot.

The hidden-localization follow-up asked whether that negative was specific to the measured site. It ran three cheap open models in parallel on Modal `L4`: `Qwen/Qwen2.5-0.5B-Instruct`, `Qwen/Qwen2.5-1.5B-Instruct`, and `HuggingFaceTB/SmolLM2-1.7B-Instruct`. The sweep preserved the same behavior conditions, then measured hidden specificity at `prompt_final`, `generated_first`, and `generated_final` token positions across early, mid, late, and final layer aliases.

This run was positive. All behavior controls passed for all three models. Behavioral bottleneck accuracy was 1.000 for SmolLM2 1.7B, 0.961 for Qwen 0.5B, and 0.938 for Qwen 1.5B. Seventeen registered hidden sites passed both the specificity and rank gates. The strongest sites were generated-final states: SmolLM2 1.7B generated-final/mid had specificity +228.301 with CI [129.438, 386.540]; Qwen 1.5B generated-final/early had +216.902 with CI [119.175, 332.137]; and Qwen 0.5B generated-final/early had +183.444 with CI [99.032, 279.601]. Qwen 1.5B also passed late/final prompt-final sites.

The interpretation is therefore narrower and stronger than the first prompt run: final behavior transfers, and hidden localization is present in the generated action trajectory. The original negative was a measurement-site failure, not evidence that prompt-level hidden geometry is absent.

The fixed-action counterfactual then tested the main confound in that positive result. For each base prompt and every slot-flipped prompt, the assistant action tokens were teacher-forced to remain fixed: either `{"tool":"noop"}` or `{"tool":"read_slot","slot":"<critical slot phrase>","value":0}`. The model therefore could not reveal the moved slot merely by generating a different answer string under the counterfactual prompt.

This stricter run was also positive. It used the same three-model `L4` grid, 192 behavior rows, and 960 hidden rows across `prompt_final`, `fixed_noop_first/final`, and `fixed_read_first/final` positions. Behavior again passed for all three models. Twenty-four registered hidden sites passed both hidden gates. Fixed noop final-layer sites passed in all three model families, with the strongest at Qwen 1.5B `fixed_noop_final/final` (specificity +12.514, CI [8.035, 18.310], rank 0.984). This removes the simple generated-answer-token explanation for the localization effect.

Finally, the causal-patch pass asked whether the localized state can shift a behavior-adjacent readout. The runner appended a partial assistant action prefix ending immediately before the JSON value token, patched one donor hidden state from the base prompt/prefix into the critical-slot-flipped prompt/prefix, and measured the donor-value minus corrupted-value next-token logit margin.

This run was positive across all three model families. Clean prompts preferred the donor value, corrupted prompts preferred the flipped value, and `value_prefix_final` late/final patch sites passed in every model. The strongest site was SmolLM2 `value_prefix_final/final`, with patch effect +7.053 and CI [6.738, 7.344]; Qwen 0.5B final had +5.503 [5.299, 5.697], and Qwen 1.5B final had +6.916 [6.375, 7.453]. Prompt-final patch sites did not pass. That negative control localizes the causal leverage to the action/value prefix state rather than the raw final prompt token.

The prompt-family robustness pass then repeated the causal-patch grid across three frozen prompt framings: the original `standard` wording, a `compact` wording, and an audit-checklist `ledger` wording. This run was also positive. All 9 prompt-family/model pairs were causally ready and patch-pass. Final-layer `value_prefix_final` effects remained large: compact

Qwen 1.5B had +7.754 with CI [7.282, 8.234], ledger Qwen 1.5B had +6.648 [6.159, 7.132], and standard SmolLM2 had +7.049 [6.751, 7.346]. Prompt-final sites remained negative controls across the robustness grid.

The API black-box evaluator then exposed the same parser-scored behavior as a public benchmark package. The command `python3 -m experiments.long_horizon_bottleneck.eval` builds cases, calls a provider adapter, writes JSONL rows, and writes a scored summary. Deterministic fixture runs validate the package locally. A real Gemini run using `gemini-3.1-flash-lite` passed the registered prompt-family behavior suite: 96 scored rows, 144 API requests, all three prompt-family cells complete, and all controls, bottleneck actions, failed-tool repairs, and successful-tool no-ops passing. A second external-stress smoke covered 20 stress/family cells over 4-slot and 8-slot settings, gap sizes 8 and 16, and two extra black-box prompt framings (`retrieval` and `dispatch`); all 20 cells passed. This result adds API-model behavioral evidence, but because the API is black-box it does not add hidden-localization or causal-patch evidence.

5. Interpretation

The core result is a **metric transport** result. The future-critical variable does not merely determine the final label; it becomes the slot to which the agent's final hidden state and commitment heads are specifically sensitive. When the critical slot moves, the sensitivity peak moves with it. When the final answer is visible at query time, the early-slot peak disappears.

The tool regimes add a second claim: the moved bottleneck survives when the agent has to externalize the critical variable through a tool interface. It must commit an argument, receive or fail to receive an external return, repair when needed, no-op when not needed, and still preserve the correct delayed decision.

This makes the diagnostic more relevant to long-horizon agents than to ordinary sequence classifiers. The bottleneck is not just "remember bit 2." It becomes "know which early variable must later be committed through an interface, recover it under feedback, and ignore matched distractors."

The prompt-level results sharpen the interpretation. Final behavior alone is not enough: a small open model can pass the prompt-level JSON behavior while one registered hidden site fails. But localization across token positions and layers recovers a positive signal, and the fixed-action counterfactual shows that the signal can survive even when generated answer tokens are held constant. The causal-patch pass then shows that the value-prefix state can shift the behavior-adjacent value-token readout, and the prompt-family pass shows that this causal leverage survives several controlled wording changes. That split is the point of the diagnostic: it can distinguish interface behavior, prompt-state memory, action-surface commitment, causal leverage over the next action token, and prompt-family robustness.

The API result adds a different kind of evidence. It shows that the same surface can be packaged as a cheap provider benchmark and that one current black-box model passes both the registered prompt-family behavior suite and a small external-validity stress suite. It does not show where the relevant state lives in that API model. The value is operational: other labs can run the same prompts and parser without needing local weights, hooks, or GPUs.

6. Boundaries

The strongest honest statement is:

> Future control relevance can move finite memory and commitment sensitivity in a synthetic long-horizon neural agent, and that moved bottleneck survives progressively harder parsed tool-interface regimes.

The result does not establish:

- production API reliability;
- autonomous language-agent tool use;
- robustness across arbitrary natural-language prompt families;
- multi-step planning in an open environment;
- human cognition or consciousness.

The latest prompt-level hidden-state result uses real pretrained tokenizers and teacher-forced/generated JSON text, but it is still a compact harness. The latest API result uses a real API model, but only as a black-box behavior surface; it does not expose hidden states or production tool execution.

7. Why This Is Valuable

The experiment's field value is not that it solves a benchmark. It gives a small, inspectable test for a pattern that current agent evaluations often miss: future relevance can reorganize internal memory geometry and action commitment.

A useful follow-on benchmark could ask, for a real agent or language model:

1. Which early facts become future-critical under a delayed objective?
2. Do internal representations or tool-call logits become specifically sensitive to those facts?
3. Does that specificity move when the objective moves?
4. Does it remain absent when the answer is visible later?
5. Does it survive tool failure, repair, and aliasing of argument names?

That is a sharper evaluation target than final-task success alone.

8. Remaining Work

The synthetic mechanism ladder is complete enough to write up and share. The prompt-level hidden-localization replication is positive, and the fixed-action counterfactual removes the simplest generated-token identity confound. The fixed-prefix causal patch also shows behavior-adjacent causal leverage at the JSON value-token readout, and the prompt-family run verifies that this leverage is not restricted to one frozen wording. The API benchmark package adds a black-box behavioral bridge and a first Gemini Flash-Lite positive. The next regimes answer different questions:

- **Multi-provider API replication:** run the same black-box suite across OpenAI, Anthropic, Gemini, and OpenAI-compatible endpoints with matched request budgets.
- **Broader external validity:** expand beyond the current small smoke to more critical slots, more episodes, longer contexts, and paraphrase families.
- **Stronger causal mediation:** patch multi-token generation or run path patching to test whether the measured state mediates the full tool action, not only the next value token.
- **LLM-agent transfer:** add real tool wrappers, multi-call traces, and API-style parser recovery around the prompt-level version.
- **Multi-step planning:** require two or more future commitments where the critical bottleneck changes after intermediate feedback.
- **Interpretability probes:** compare the hidden-state metric to attention, activation patching, or linear probes over the critical slot.

The most valuable immediate next step is to publish the diagnostic as a compact benchmark card: synthetic ladder positive, prompt-level behavior positive, prompt-level hidden localization positive, and fixed-action localization positive after generated action tokens are held constant, with a first causal patch result showing value-token logit control, prompt-family robustness, and a black-box API benchmark surface.

9. Reproducibility

Experiment code lives in `experiments/long_horizon_bottleneck/`.

Synthetic ladder runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/long_horizon_bottleneck/modal_stochastic_tool_failure_sweep.py \
    --seeds 4 --train-steps 900 --architectures transformer \
    --conditions autoregressive_json_bottleneck,autoregressive_json_visible_control \
```

```
    --critical-slots 0,1,2,3 \
    --aliases-per-slot 3 \
    --failure-probability 0.5 \
    --budget-usd 25 \
    --out artifacts/long_horizon_bottleneck/autoregressive_json_surface_l4.json
```

Prompt-level confirmatory runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/long_horizon_bottleneck/modal_prompt_json_transfer_sweep.py \
    --model-id Qwen/Qwen2.5-0.5B-Instruct \
    --seeds 4 \
    --episodes-per-cell 8 \
    --hidden-metric-episodes 2 \
    --critical-slots 0,1,2,3 \
    --budget-usd 25 \
    --base-seed 20260800 \
    --out artifacts/long_horizon_bottleneck/prompt_json_transfer_l4.json
```

Prompt-level hidden-localization runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/long_horizon_bottleneck/modal_prompt_json_hidden_localization_sweep.py \
    --models Qwen/Qwen2.5-0.5B-Instruct,Qwen/Qwen2.5-1.5B-Instruct,HuggingFaceTB/SmolLM2-1.7B-
Instruct \
    --seeds 4 \
    --episodes-per-cell 8 \
    --hidden-metric-episodes 2 \
    --critical-slots 0,1,2,3 \
    --hidden-positions prompt_final,generated_first,generated_final \
    --hidden-layers early,mid,late,final \
    --budget-usd 25 \
    --base-seed 20260850 \
    --out artifacts/long_horizon_bottleneck/prompt_json_hidden_localization_l4.json
```

Prompt-level fixed-action localization runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/long_horizon_bottleneck/modal_prompt_json_hidden_localization_sweep.py \
    --models Qwen/Qwen2.5-0.5B-Instruct,Qwen/Qwen2.5-1.5B-Instruct,HuggingFaceTB/SmolLM2-1.7B-
Instruct \
    --seeds 4 \
    --episodes-per-cell 8 \
    --hidden-metric-episodes 2 \
    --critical-slots 0,1,2,3 \
    --hidden-positions
prompt_final,fixed_noop_first,fixed_noop_final,fixed_read_first,fixed_read_final \
    --hidden-layers early,mid,late,final \
    --budget-usd 25 \
    --base-seed 20260900 \
    --out artifacts/long_horizon_bottleneck/prompt_json_fixed_action_localization_l4.json
```

Prompt-level fixed-prefix causal patch runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/long_horizon_bottleneck/modal_prompt_json_causal_patch_sweep.py \
    --models Qwen/Qwen2.5-0.5B-Instruct,Qwen/Qwen2.5-1.5B-Instruct,HuggingFaceTB/SmolLM2-1.7B-
Instruct \
    --seeds 4 \
    --episodes-per-cell 8 \
    --critical-slots 0,1,2,3 \
    --patch-positions prompt_final,value_prefix_final \
    --patch-layers late,final \
    --budget-usd 25 \
    --base-seed 20260950 \
    --out artifacts/long_horizon_bottleneck/prompt_json_causal_patch_l4.json
```

Prompt-level prompt-family causal patch runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    uvx --python 3.12 --from modal modal run \
    experiments/long_horizon_bottleneck/modal_prompt_json_causal_patch_sweep.py \
    --models Qwen/Qwen2.5-0.5B-Instruct,Qwen/Qwen2.5-1.5B-Instruct,HuggingFaceTB/SmolLM2-1.7B-
Instruct \
    --prompt-families standard,compact,ledger \
    --seeds 4 \
    --episodes-per-cell 8 \
    --critical-slots 0,1,2,3 \
    --patch-positions prompt_final,value_prefix_final \
    --patch-layers late,final \
    --budget-usd 25 \
    --base-seed 20261000 \
    --out artifacts/long_horizon_bottleneck/prompt_json_prompt_family_causal_patch_l4.json
```

API black-box prompt-family runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    env PYTHONPATH=. python3 -m experiments.long_horizon_bottleneck.eval \
    --provider gemini \
    --models gemini-3.1-flash-lite \
    --suite prompt_family \
    --prompt-families standard,compact,ledger \
    --seeds 1 \
    --episodes-per-cell 2 \
    --critical-slots 0,1,2,3 \
    --max-requests 150 \
    --max-output-tokens 64 \
    --sleep-seconds 0.05 \
    --out artifacts/long_horizon_bottleneck/
api_blackbox_gemini_flash_lite_prompt_family_summary.json \
    --jsonl artifacts/long_horizon_bottleneck/
api_blackbox_gemini_flash_lite_prompt_family_rows.jsonl
```

API black-box external-stress runner:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
    env PYTHONPATH=. python3 -m experiments.long_horizon_bottleneck.eval \
    --provider gemini \
    --models gemini-3.1-flash-lite \
    --suite external_stress \
    --prompt-families standard,compact,ledger,retrieval,dispatch \
```

```
    --seeds 1 \
    --episodes-per-cell 1 \
    --critical-slots 0 \
    --n-slots 4,8 \
    --slot-gap 8,16 \
    --max-requests 150 \
    --max-output-tokens 64 \
    --sleep-seconds 0.05 \
    --out artifacts/long_horizon_bottleneck/
api_blackbox_gemini_flash_lite_external_stress_summary.json \
    --jsonl artifacts/long_horizon_bottleneck/
api_blackbox_gemini_flash_lite_external_stress_rows.jsonl
```

Local verification for the code paths:

```
uvx ruff check experiments/long_horizon_bottleneck tests/test_long_horizon_bottleneck.py
uvx ty check experiments/long_horizon_bottleneck tests/test_long_horizon_bottleneck.py
python -m compileall -q experiments/long_horizon_bottleneck tests/test_long_horizon_bottleneck.py
python -m pytest tests/test_long_horizon_bottleneck.py -q
```

The raw Modal artifacts are kept under gitignored `artifacts/`; committed result reports in `experiments/long_horizon_bottleneck/results/` summarize every run.